

When Low-Cost Language Models Cooperate by Default: Incentives, Partners, and Selection in a Collective-Risk Dilemma

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Abstract

Language-model agents increasingly act in shared environments where one local choice can expose a whole group to collective loss. We study whether cooperation follows the stakes, the partners, or an unspoken default. In a collective-risk social dilemma, agents contribute to a shared threshold over repeated rounds; a catastrophe destroys savings when the threshold is missed. We evaluate five low-cost production models through self-play, prompt controls, value questions, scripted partners, mixed tables, and payoff-based selection. The main result is a dissociation between cooperative-looking play and incentive response. Four models contribute near the equal share at every positive risk level, and most also pay at zero risk, where contribution cannot help. Removing normative wording leaves this pattern largely intact, whereas stating that only the agent’s own cash matters reduces zero-risk payment for most models. Several models answer risk comparisons correctly in separate questions but barely change their actions. Against fixed partners, no model consistently follows the exact best response, and some pay after the outcome is settled. In mixed tables, one late under-contributor can break an otherwise successful group. Under rare-mutation payoff-based imitation, a population of one table favours the smallest payer and harms group success. At moderate and high risk, a larger randomly mixed population instead favours an exact fair-share policy. These selection results are conditional on the base-prompt payoffs. These findings show why self-play success is not enough to evaluate multiagent systems. Evaluations should state the agent’s objective, compare behaviour with a computed optimum, test situations where actions are dominated, and measure performance in mixed populations.

Keywords

LLM agents, collective-risk social dilemma, cooperation, best response, empirical game-theoretic analysis, multiagent evaluation

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1 Introduction

Language-model agents are starting to act for people in shared settings: they spend budgets, share compute and bargain over common resources. Each such agent is a player in a social dilemma, because

what it pays helps the group and costs its owner. Before deploying one, a designer needs to know two things. Does the agent’s cooperation depend on what is at stake? And does it depend on what the other agents do?

Self-play, a common test, cannot answer either question. When every seat holds a copy of the same model, an agent that reasons about the stakes and an agent that repeats a fixed rule can leave the same record of play. A success rate also hides how far the agents were from the best they could have done: a group can reach its goal and still waste most of its money. And a prompt that gives a group target and a number of rounds already suggests an equal split, so an agent that simply repeats that split looks cooperative on every summary measure.

We study both questions in a classic game from research on climate cooperation, the collective-risk social dilemma [17]. Players pay into a shared pool over several rounds, and if the pool misses a target, a catastrophe destroys everyone’s savings with probability p . The group-welfare benchmark has a simple shape. For risk-neutral players, it is to pay nothing below risk one half and exactly reach the target above it. One symmetric optimum has everyone pay the *fair share*, the target split equally. When there is no risk at all, paying only lowers a player’s own money, so risk attitude alone cannot explain it.

We use *default* as a behavioural label for near-flat payment across risk levels under a fixed prompt, not as a claim about internal reasoning or invariance to partners. Payment at zero risk and after settlement tests whether this pattern extends to actions that cannot improve cash payoffs. We test five low-cost production models: Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna, Qwen3-235B and Grok 4.20. Four designs combine a full self-play risk grid, prompt controls and value questions, scripted partners with computable best responses, and real mixed tables that supply payoffs for selection [31, 33]. Our contribution is this linked evaluation, not a new equilibrium or selection principle. We report four findings (Figure 1).

- **Risk does not change the default** (Section 4). Most models pay about the fair share at every positive risk level, and most also pay when there is no risk. Removing normative words changes neither result. Stating that only the agent’s own cash matters cuts most zero-risk payment, but only GPT-5.6 Luna then moves from paying little at low risk to the fair share at high risk.
- **Answers track the risk; play does not** (Section 5). Asked in separate calls, several models rank the two choices correctly and change that ranking with the risk, but they keep paying the same amount. Beside scripted partners who make every payment a loss, no model keeps all its money, and two models keep paying after the outcome is settled.
- **An exact fair share has no slack** (Section 6). One partner that stops paying in the last round often breaks a table of



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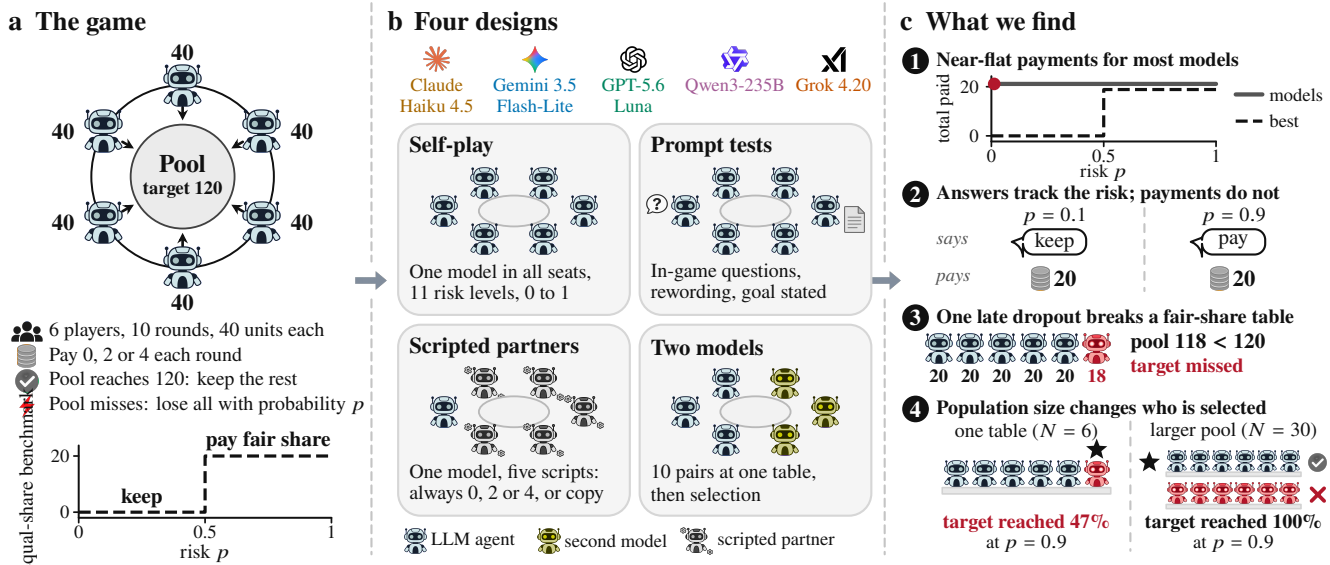


Figure 1: Overview. (a) Six players hold 40 units and pay 0, 2 or 4 per round for ten rounds; if the pool misses 120, all savings are lost with probability p . Below: symmetric welfare benchmark (Proposition 1; both 0 and 20 are optimal at $p = 0.5$). (b) Models and designs. (c) Findings of Sections 4 to 7; its target rates are self-play rates at $p = 0.9$ weighted by stationary mass, not the success rates of the depicted mixed tables.

exact fair-share players, while a table that overpays absorbs it. Most agents pay less when generous partners fill most of the table.

- **Population size changes who is selected** (Section 7). Under payoff-based imitation, a population of one table favours the smallest payer and lowers group success. In a larger randomly mixed population, the same process favours the exact fair-share player at moderate and high risk.

For the models we test, payment is a default rather than a response to the stakes. A default can work well beside partners like itself, and how a system states its agents' goals and selects its agents can make its failures better or worse.

2 Related work

Collective risk and selection. The collective-risk dilemma was built to study why groups fail to avoid a shared disaster [17]. Human groups reach the target mostly when the risk is high [17], and inequality and communication change the result [27]. Games with a threshold have many equilibria [19], human contributions decline over the rounds of repeated public goods games [2], and the risk of collective failure can turn a defection trap into a coordination problem [22]. Theory also shows that selection on payoff relative to a small group favours whoever does better than its neighbours [23], while selection between groups can favour cooperation [30]. This literature tells us that behaviour *should* move with the risk, so a flat response is informative, and we reuse its selection theory to read our mixed tables.

Language models in games. Language models have been studied as economic subjects [4, 8, 16], in repeated two-player games [1, 9], and in multi-player game benchmarks, where contributions to a

public good can stay the same even when it returns nothing [11]. Models tend to be more cooperative than people [16], an even split can reflect a focal point rather than fairness [6], and models can hold a correct belief or plan and act against it [7, 24]. In social dilemmas, societies of agents over-use a common resource [21], reasoning models free-ride in public goods games [10], and agents stand in for people in a collective-risk game [25]. A study of 35 open models includes a collective-risk game at one risk level and finds play far from the group optimum [13]. Willis et al. study a binary collective-risk game with strategies written by three models and find that payoff-biased imitation drives larger groups to selfish, low-welfare outcomes [34]. In another study, prompts that evolve under individual selection collapse into defection while group selection keeps them prosocial [3]. We differ in three ways. We vary the risk from zero to certainty and compare turn-by-turn play with a computed optimum, adding two tests where paying cannot help. We compute the exact best response to fixed partners. And we select among deployed models from their real mixed tables, showing that the switch from tablemate to across-table scoring happens at seven or eight agents.

Multiagent evaluation. Multiagent learning and cooperative AI ask for agents that work with partners unlike themselves [5, 12, 14, 15, 26]. Empirical game-theoretic analysis treats trained agents as strategies of a meta-game [31, 33], and α -Rank ranks them by evolutionary dynamics [18]. Tournaments that mix language models with classic strategies report model-specific strategic fingerprints in the prisoner’s dilemma [20], cooperation evolves differently across base models in a donor game [32], and cooperation sustained by repetition weakens when co-players differ [29]. In theory, adaptive

Table 1: Symbols and terms.

Term	Meaning
Seat; table	One player position; the six seats of one game
Fair share	The target split equally: 20 per seat, or 2 per round
Exact split	Paying 2 in every one of the ten rounds
Pivot $p^* = \frac{1}{2}$	Risk above which reaching the target pays
$\text{opt}(p)$	Welfare-optimal expected payoff per seat
Dominated	Lowers the seat’s own payoff whatever others do
Settled	The history proves that more payment cannot change the outcome: the pool has reached T , or cannot reach it even if every seat pays 4 from now on

players in a collective-risk game compensate for the effort of artificial agents [28]. We add a many-player dilemma in which the best response to fixed partners can be computed exactly, and measure this compensation with real models.

3 Game-theoretic benchmark and evaluation design

3.1 Rules

Six players each start with $e = 40$ units, and the game lasts $R = 10$ rounds. In each round every player privately chooses a contribution $a_i^t \in \{0, 2, 4\}$ to a shared *pool*, and after each round all players see what every seat has paid. Let $x_i = \sum_t a_i^t$ be a player’s total and $X = \sum_i x_i$ the pool. The target is $T = 120$. A player keeps what it did not pay, but if the pool misses the target, a catastrophe destroys all savings with probability p (Table 1 lists the terms we use):

$$u_i = \begin{cases} e - x_i & \text{if } X \geq T, \\ 0 \text{ with probability } p, \text{ else } e - x_i & \text{if } X < T. \end{cases} \quad (1)$$

Proposition 1. *Consider the expected-payoff game in which each player picks a total $x_i \in \{0, 2, \dots, 40\}$, and let $p < 1$. A profile is a Nash equilibrium if and only if $x = 0$, or $X = T$ and $x_i \leq pe$ for every i . A target-reaching equilibrium exists if and only if $p \geq p^* = T/(6e) = \frac{1}{2}$. The welfare-optimal outcome is $x = 0$ for $p < p^*$ and $X = T$ for $p > p^*$, and both are optimal at $p = p^*$, so the welfare-optimal mean payoff per player is $\text{opt}(p) = \max\{(1-p)e, e - T/6\}$; this welfare statement also holds at $p = 1$.*

Proof sketch. If $X < T$, any player with $x_i > 0$ gains by paying nothing, so the only equilibrium that misses the target is $x = 0$. If $X > T$, then $X \geq T + 2$ because totals are even, so a player can pay 2 less and still reach the target. If $X = T$, the best deviation is to pay nothing, worth $(1-p)e$ against $e - x_i$, which gives $x_i \leq pe$; summing over players gives $p \geq T/(6e)$, and conversely $x_i = 20$ for every i is an equilibrium when $p \geq \frac{1}{2}$. Welfare is $6e - X$ when the target is met and $(1-p)(6e - X)$ otherwise, so it peaks at $X = T$ or $X = 0$. The ten-round game, in which strategies can react to the history, has the same totals in pure-strategy Nash equilibrium (full proof in the supplementary material); we claim nothing about subgame perfection, and $\text{opt}(p)$ is a welfare bound that does not depend on the equilibrium concept. $\square$

What we compare play with. Proposition 1 gives a *symmetric welfare benchmark*: keep everything below the pivot and pay the fair

Table 2: Designs. A cell is one condition at one risk level: one model in self-play, one model with one kind of scripted partner, or one pair of models with one split of seats. Each cell holds ten games.

Design	Risk levels p	Games
Self-play grid	0, 0.1, ..., 1	550
In-game questions	0.1, 0.5, 0.9	150
No hint	0.1, 0.5, 0.9	150
Neutral wording	0, 0.1, 0.9	150
Neutral wording, goal stated	0, 0.1, 0.9	150
Two rewodings	0.1, 0.9	200
Temperature zero	0.1, 0.9	100
One model, five scripts (4 kinds)	0.1, 0.3, ..., 0.9	1,000
Two models (10 pairs, 1 to 5 seats)	0.1, 0.5, 0.9	1,500
All		3,950

share above it. Asymmetric allocations reaching exactly T attain the same group welfare. This is not a unique equilibrium or a prediction of individual play: paying nothing is also an equilibrium at every risk level. Paying is dominated in own cash at $p = 0$ and after the outcome is settled, provided a missed target still leaves a positive chance of keeping savings ($p < 1$). These tests do not require risk neutrality or beliefs about the partners. They do assume that more own cash is preferred; non-cash goals are not ruled out.

Proposition 2. *For two seats i and j in the same game, the difference in their expected payoffs over the lottery is $c(x_j - x_i)$, with $c = 1$ if the target was reached and $c = 1 - p$ otherwise.*

Both seats face the same pool and the same lottery, so only their own payments differ. Hence, for $p < 1$, whoever pays less at a table earns more, whatever the group achieves.

3.2 Models, designs and measures

We test five production models, one per provider, all from the low-cost tier of their family: Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna, Qwen3-235B and Grok 4.20. We chose this tier because agents that act at scale are likely to run on it. Qwen3-235B is the instruction-tuned build without a thinking mode, and Grok 4.20 is its non-reasoning variant. Table 2 lists the designs, and every model plays the same number of games in each of them.

Prompts. In each round a seat receives one prompt with the rules, the full history of every seat’s payments, and a request for one contribution; the supplementary material gives every prompt word for word. The base prompt calls the pool a “climate account”, says that the group “must” reach the target, calls the loss a “disaster” and names the game as a collective-risk social dilemma. It never states the player’s goal, and it hints at the equal split twice. The prompt controls change exactly these parts. *No hint* deletes the two equal-split phrases. *Neutral wording* replaces the normative and naming phrases with plain ones, such as “a shared account”. *Goal stated* keeps the neutral wording and adds an opening that says the player is in a paid experiment where “the only thing that matters to you is your own final cash payoff”. Two *rewodings* change the phrasing but keep every rule and the hint.

Protocol. The running pool is not printed, so a seat that wants it must add up the history. The prompt is in English, no seat is given a persona, and the temperature is 0.7 unless stated. We read the move from the reply’s final CONTRIBUTION: line. A reply that is cut off at the output limit of 3,000 tokens is asked again with a larger limit rather than guessed, and all but 2 of the 187,000 replies could be read.

Measures. We report *expected* payoffs, the average a seat would earn if the catastrophe lottery were run many times on the same payments, and the share of $\text{opt}(p)$ that is lost. Proposition 1, $\text{opt}(p)$ and the selection fitness below are all defined in expected payoffs, and given the payments the expected payoff removes the lottery noise without bias. Agents never see the lottery draw, and the supplementary material lists the realised draws. The unit of analysis is the game, since seats in one game share a history and a lottery. Intervals are 95% percentile bootstraps over games, and tests are permutation tests over games. The supplementary material gives every value behind the statements below.

4 Risk does not change the default

Four of the five models pay about the fair share at every positive risk level (Figure 2a). Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna and Qwen3-235B pay almost the same amount below the pivot as above it (a step of at most 0.2 units), whereas the symmetric welfare benchmark jumps from nothing to the fair share. Human groups with the same rules reach the target in half of their games at $p = 0.9$ and in none at $p = 0.1$ [17]. Strong risk aversion could explain a flat line above zero risk, since a catastrophe leaves nothing and a player with relative risk aversion near one already prefers a certain fair share at low risk. Risk attitude alone cannot explain paying at zero risk, nor can concern for others’ cash, since that payment helps nobody. Gemini 3.5 Flash-Lite is the clearest case. Almost every seat plays the exact split at every risk level (Table 3), so a near-fixed rule attains the welfare optimum above the pivot but not below it.

Paying when there is nothing to protect. Four of the five models also pay at $p = 0$, about as much as at any other risk level, although paying there only lowers a player’s own money (Table 3, first column). GPT-5.6 Luna is the exception. It pays nothing at $p = 0$ and its usual share at every positive risk, so it tells no risk apart from some risk but does not respond to how large the risk is.

Three ways to lose. The models lose payoff in three different ways (Table 3, Figure 2c). Gemini 3.5 Flash-Lite, Claude Haiku 4.5 and GPT-5.6 Luna lose mostly at low risk, where they pay for a target that is not worth reaching. Grok 4.20 overpays: its groups reach the target well before the last round, and its seats keep paying into a pool that is already safe. Qwen3-235B fails at the end. It pays 2 round after round, then almost all of its seats pay nothing in the last round (Figure 2b), so its groups almost never reach the target; had each seat simply repeated its previous move, every one of those groups would have succeeded. The skip follows the clock, not the pool, much as human contributions decline at the end of repeated public goods games [2]: across all games with the base prompt, Qwen3-235B skips the last payment as often when its payment is needed as when the others’ fair shares already suffice (93% of seats

Table 3: Self-play over the risk grid. Pays at $p=0$: mean seat total where paying is dominated. Exact split: seats paying 2 in all rounds. Target: games reaching it. Payoff: expected payoff per seat; the optimum averages 25.5. Lost: share of $\text{opt}(p)$ lost. Main loss: paying below p^* , paying past 120, or missing the target at or above p^* (in the last round).

Model	Pays at $p=0$	Exact split (%)	Target (%)	Payoff	Lost Main (%)
Claude Haiku 4.5	20.9	17	98	18.9	26 below p^*
Gemini 3.5 Flash-Lite	20.0	99.7	100	20.0	21 below p^*
GPT-5.6 Luna	0.0	72	70	19.2	25 below p^*
Qwen3-235B	18.0	4	2	10.9	57 last round
Grok 4.20	37.4	2	100	5.0	80 overshoot

in both cases), and it almost never stops in an earlier round once the target is met. GPT-5.6 Luna fails in the opposite way: its last move depends on whether its payment is needed. It skips in 2% of cases when the target needs its payment, but in 53% when the others’ fair shares would suffice. When six copies of this rule each count on another seat to pay, tables that arrive ahead of the fair pace of 12 per round often miss the target, while tables exactly on pace reach it. Claude Haiku 4.5 shows that the pool can be read from the same prompt, since its seats stop paying in the last round mostly when the target has already been reached. Success rates hide all of this. Gemini 3.5 Flash-Lite and Grok 4.20 both reach the target in every game, yet Grok 4.20 keeps far less of its money. For $0.5 < p < 1$, Qwen3-235B and Grok 4.20 groups earn less in most games than in the worst equilibrium of Proposition 1, in which nobody pays.

The default survives rewording, but its failures do not. Deleting the two equal-split phrases lowers no model’s payment (Table 4), so the hint set a ceiling and guided precision but was not the source of the cooperation. Gemini 3.5 Flash-Lite still plays the exact split, which it can rebuild from the target, while GPT-5.6 Luna keeps its average payment but plays the exact split less often. Qwen3-235B and Grok 4.20 pay more without the hint; Qwen3-235B now reaches the target, but it earns less because it overpays. The two rewordings and a temperature of zero leave the payments of the four models other than Grok 4.20 almost unchanged. The failures are less stable than the averages. Under the second rewording Grok 4.20 pays about the fair share, and how often Qwen3-235B drops out in the last round swings widely from one prompt to another. Grok 4.20’s small rise with risk on the grid mostly disappears in the rerun, so we do not read it as a response to risk.

Stating the goal, not removing the norms, cuts payment at zero risk. The default does not come from the normative words. With neutral wording, most models still pay at $p = 0$, and at low and high risk four models pay on average what they paid before, although GPT-5.6 Luna plays the exact split less often, while Grok 4.20 pays even more (Table 4). Adding the stated goal is what changes play at zero risk: Claude Haiku 4.5 and Gemini 3.5 Flash-Lite then pay almost nothing there, and Qwen3-235B pays far less than before, while Grok 4.20 barely changes. The goal still does not make most models weigh the risk. Claude Haiku 4.5 and Gemini 3.5 Flash-Lite pay the fair share already at low risk, so they tell no risk apart from some

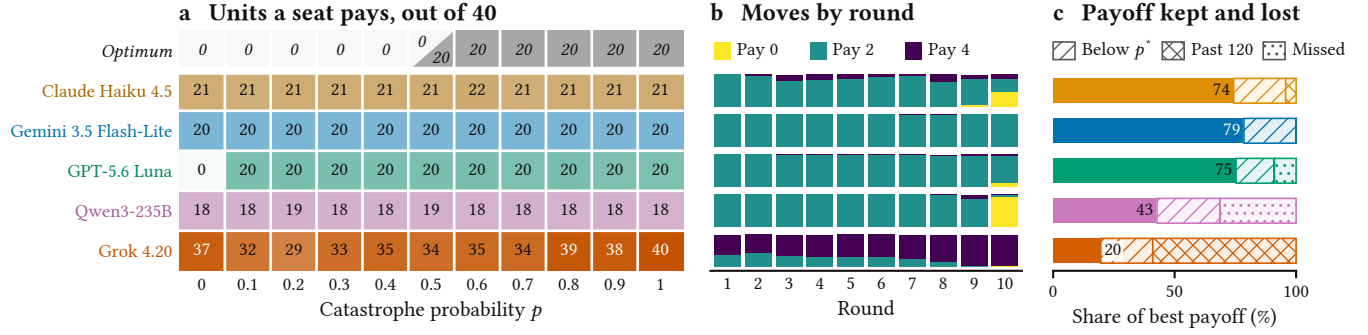


Figure 2: Self-play on the risk grid, 110 games per model. (a) Mean units a seat pays out of 40, ten games per cell; the top row is the optimum of Proposition 1, where both 0 and 20 are optimal at $p = 0.5$. (b) Share of seats paying 0, 2 or 4 in each round, over the games with $p > 0$. (c) Expected payoff kept as a share of $\text{opt}(p)$, and the loss split into paying toward the target below p^* , paying past 120, and missing the target at or above p^* .

Table 4: Mean seat total under prompt and decoding changes. Top: pooled over $p = 0.1$ and 0.9 (20 games each). Rerun repeats the base prompt with in-game questions; Para. 1, 2 are the rewordings; Wording is the neutral wording and Goal adds the stated goal. Bold: a shift from Base of at least 2 units with permutation $P < 0.01$, every bold shift has $P < 0.001$, below the Bonferroni threshold for the 35 comparisons. Bottom: single risk levels (10 games each).

Condition	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20
Base	20.9	20.0	19.9	18.4	35.1
Rerun	21.3	20.0	19.9	18.5	35.6
No hint	21.7	20.0	19.9	34.7	40.0
Para. 1	21.3	20.0	19.9	19.4	31.8
Para. 2	20.9	20.0	19.8	18.0	20.6
Temp. 0	21.2	20.0	20.0	18.3	35.1
Wording	21.0	20.0	19.4	18.7	39.7
Goal	20.0	20.1	13.9	8.2	37.7
Base $p=0$	20.9	20.0	0.0	18.0	37.4
Wording $p=0$	18.0	19.9	0.4	18.1	39.8
Goal $p=0$	0.3	1.4	0.2	4.8	36.7
Goal $p=0.1$	19.8	20.2	7.7	6.4	36.7
Goal $p=0.9$	20.2	20.0	20.1	10.1	38.7

risk but do not grade it, a pattern that strong risk aversion could also produce. Only GPT-5.6 Luna moves from paying little at low risk to paying the fair share at high risk. Qwen3-235B pays somewhat more at high risk than at low risk, but less than before at every risk level, and never reaches the target; Grok 4.20 keeps paying far more than needed. Payment at zero risk therefore depends on leaving the goal unstated for most models, while the flat response above zero risk does not. The scripted and mixed designs below use the base prompt and describe this default.

5 Answers track the risk; play does not

Perhaps the agents simply do not understand the game. We ask the playing agent which choice pays more, and we seat it beside fixed

partners, against whom the *best response*, the play that earns the most, can be computed exactly.

5.1 Asking the agent

In the in-game question games we question the first seat at the start, in the middle and at the end of the game, about the state it has just decided on. Each question is a separate call, and its answer never enters the game. Rule questions only ask for numbers printed in the prompt, and all are answered correctly. We therefore rest on one value question, asked word for word:

Compare two strategies for the whole game. (A) You contribute 2 in every one of the 10 rounds and the group reaches the target, so you finish with 20 for certain. (B) You contribute 0 in every round and the group does not reach the target, so you keep 40 unless the disaster strikes. Which gives the higher AVERAGE final cash? Answer 1 if A is higher, 2 if B is higher, 0 if they are exactly equal.

The correct answer is B below p^* , A above it, and equal at p^* . A second value question could be read in two ways, so we report it only in the supplementary material.

Answers move with the risk; play does not. Claude Haiku 4.5 and Gemini 3.5 Flash-Lite answer the value question correctly almost every time, and GPT-5.6 Luna does so most of the time. All three change their answer with the risk (Figure 3a): at low risk they almost always say that keeping pays more, and at high risk they mostly say the opposite. Their payments barely change (Figure 3b). Qwen3-235B always says that keeping pays more, whatever the risk, and Grok 4.20 gives that answer in the first round whatever the risk. The question supplies two hypothetical outcomes, not the partners' responses to a deviation, so correct answers do not establish knowledge of a best response. It also makes the comparison explicit, unlike the move request. The result is therefore a gap between answers and play, not evidence about what an agent computes when deciding. A strongly risk-averse agent could give these answers and still pay; payment at $p = 0$ and after settlement provides the sharper test.

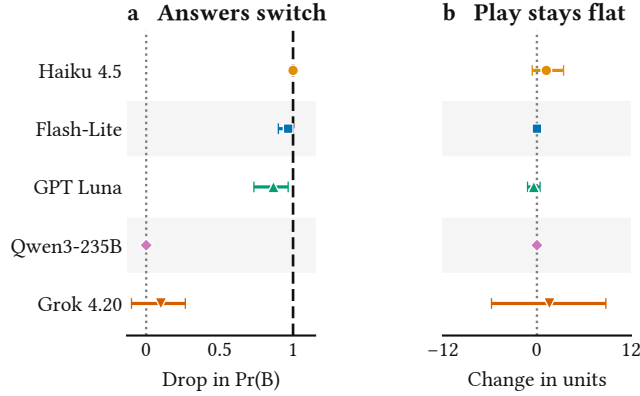


Figure 3: Answers and play as risk rises from 0.1 to 0.9 (ten games per model and risk, 95% bootstrap intervals). (a) Drop in the share of answers favouring B; dashed: a fully correct switch. (b) Change in the questioned seat’s contribution. Dotted: no change. This compares answers and actions, not best responses; all three risks are detailed in the supplement.

5.2 Playing against fixed partners

Next we seat one model beside five scripted partners of one kind. They always pay 0, always pay 2, always pay 4, or copy the group, which means paying 2 in the first round and then the rounded average of what the other seats paid in the previous round. The model is not told that its partners are scripted, but their moves are in the history. We found the best response to each kind by trying every possible sequence of the model’s own moves. Beside partners who always pay 0 the target cannot be reached, and beside partners who always pay 4 it is reached without the model, so the best response to both is to pay nothing; once the history settles the outcome, paying is also dominated. Beside fair-share partners the best response is to pay nothing below p^* and exactly the fair share above it, and partners who copy the group never move away from 2 and give the same answer.

No model keeps all its money where that is the best response. In no game beside partners who always pay 0 or always pay 4 does the model pay nothing. Only 186 of the 1,000 scripted games are exactly optimal, and every one of them is a game in which the optimum is the fair share. The models land on the optimum only where the optimum is the default. An agent that does not know its partners are scripted may reasonably pay early to learn who they are, so the cleaner evidence is payment after the outcome is settled (Table 5).

Partners matter; risk does not. We split the variation in what a model pays beside the three fixed-amount partners into parts due to the partner, the risk and their interaction. For Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna and Qwen3-235B, the kind of partner explains most of the variation (69% to 92%) and the risk with its interaction at most 5%, while for an exact best responder in the same games the risk would explain half of it. Beside fair-share partners, where the best response jumps from nothing to the fair share at p^* , no model’s total moves by more than a single payment of 2 between low and high risk. Grok 4.20’s total follows neither: it

Table 5: Units a model pays per game before and after the outcome is settled, beside five partners who always pay 0 or always pay 4 (50 games per entry). Given these partners, every unit paid is wasted; units paid after settlement are wasted even for an agent that does not know its partners are scripted.

Model	Always 0		Always 4	
	Before	After	Before	After
Claude Haiku 4.5	7.7	0.1	14.2	1.1
Gemini 3.5 Flash-Lite	13.6	0.2	10.9	0.0
GPT-5.6 Luna	8.5	0.0	4.5	0.0
Qwen3-235B	17.8	12.1	12.3	8.1
Grok 4.20	19.6	11.1	16.7	16.2

is set mostly by its opening move, made before any history exists, which is the largest payment in about half of its games and which it mostly repeats.

Responses to partners take simple forms. The models respond to their partners, but not the way a best responder would (Table 5; the supplementary material shows every round). Gemini 3.5 Flash-Lite pays its usual rate, and it and Claude Haiku 4.5 stop once the history shows that the outcome is settled. GPT-5.6 Luna pays little even before that point beside partners who always pay 4, which comes close to a best response while it learns who its partners are. Qwen3-235B pays more, not less, beside partners who pay nothing. Qwen3-235B and Grok 4.20 keep paying large amounts after the outcome is settled, which no belief about the partners can justify. Qwen3-235B’s last-round rule also costs its partners. Beside five fair-share partners it pays 2 for nine rounds and then nothing, so the table misses the target in almost every game, each time by that single missing payment. Gemini 3.5 Flash-Lite in the same seat reaches the target every time.

6 Defaults in mixed groups

To see what happens when agents with different defaults share a table, we seat every pair of models together, with one to five seats for the first model, at low, moderate and high risk. The seats of each model form one block whose position is fixed within a cell and alternates across cells, so every model holds every seat equally often overall.

At a shared table every model earns more than Grok 4.20, while the payoff gaps among the other four models are small (Figure 4, above the diagonal). This follows from Proposition 2: Grok 4.20 pays the most, and at one table the larger payer earns less. Group success depends on the pair (Figure 4, below the diagonal). Tables that include Grok 4.20 reach the target in the great majority of games, while tables that mix Qwen3-235B with Gemini 3.5 Flash-Lite or GPT-5.6 Luna rarely do.

An exact fair share has no slack. One partner that stops paying in the last round can break a table of exact fair-share players (Figure 5a). When the pool is one round of fair shares short before the last round, Qwen3-235B pays nothing in 92% of cases and Gemini 3.5 Flash-Lite in 0.1%. Five Gemini 3.5 Flash-Lite seats with

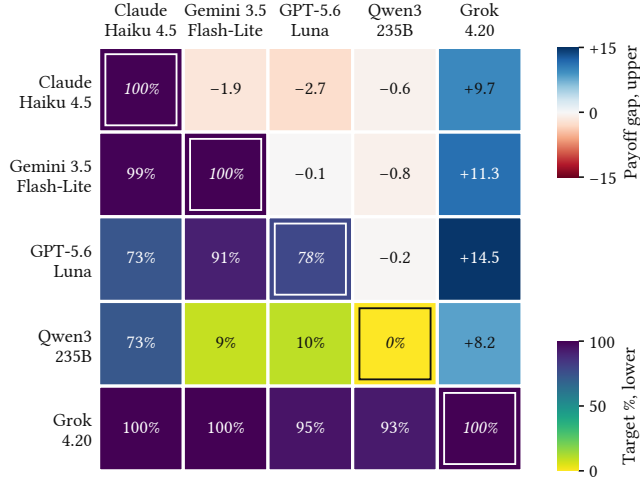


Figure 4: Two models at one table, 150 games per pair, pooled over seat splits and risk. Above the diagonal: expected payoff of the row model’s seats minus the column model’s, in the same games. On and below it: share reaching the target; boxed: self-play (60 games, grid plus question games at the same risks).

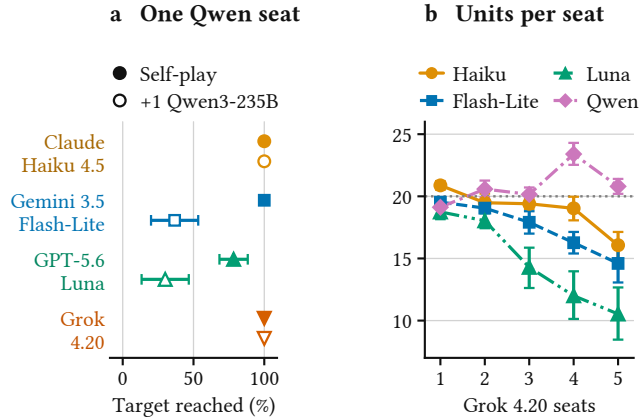


Figure 5: Mixed groups, pooled over $p = 0.1, 0.5, 0.9$. (a) Self-play (filled; 60 games, grid plus question games) versus one Qwen3-235B replacement (open; 30). (b) Mean contribution per focal seat beside one to five Grok 4.20 seats (30 games each); dotted: fair share. Bars: 95% game-bootstrap intervals, pooled without risk stratification. Lines join measured means.

one Qwen3-235B seat succeed in only 11 of 30 games, although Gemini 3.5 Flash-Lite alone never misses in our sample; GPT-5.6 Luna also loses success. Tables of Claude Haiku 4.5 or Grok 4.20, which pay more, absorb the same partner. Replacing *all* Qwen3-235B last-round zeros by 2, holding other recorded actions fixed, would rescue 92% of the 323 failed mixed tables containing it. This is an arithmetic repair, not a tested intervention. Conversely, one Grok 4.20 seat lets five Qwen3-235B seats succeed in most games

and pays for the rescue. These comparisons depend on the base prompt and the fixed seat blocks.

Generous partners are free-ridden on. Most agents pay less as the number of Grok 4.20 seats rises from one to five (Figure 5b). Beside five, GPT-5.6 Luna pays 10.5 units; Gemini 3.5 Flash-Lite and Claude Haiku 4.5 also fall below the fair share. The latter two stop once the pool reaches the target, while GPT-5.6 Luna pays less from the start. Qwen3-235B does not show this decline. Within a cell, Grok 4.20’s higher opening payments are associated with lower total payments by Gemini 3.5 Flash-Lite and Claude Haiku 4.5, mainly through earlier target completion. GPT-5.6 Luna also pays less while the pool is short, consistent with the compensation predicted for adaptive players beside artificial agents [28]. Openings are not randomised, and partner behaviour can change after a deviation, so this association does not identify a causal response or an exact best response.

7 Which agent wins under selection

Systems that host many agents may retain or copy agents with high payoffs. We ask how population size changes this process, treating each model as a strategy in an empirical game [31, 33]. We write $\pi_A(k)$ for the mean expected payoff of an A seat at a table with k seats of A and $6 - k$ of B , estimated from mixed tables for $1 \leq k \leq 5$ and self-play for $k = 6$.

Selection process. A population of N agents holds i agents of type A and $N - i$ of type B , and each agent’s five tablemates are drawn at random from the other $N - 1$, as in [22]. The fitness of each type is its payoff averaged over the tables it can land in:

$$f_A(i) = \sum_{k=1}^6 \mathcal{H}(k-1; N-1, i-1) \pi_A(k), \quad (2)$$

$$f_B(i) = \sum_{k=0}^5 \mathcal{H}(k; N-1, i) \pi_B(6-k),$$

where $\mathcal{H}(k; N-1, m)$ is the hypergeometric chance that k of five tablemates come from the m other agents of type A . Under Fermi imitation, an agent copies a random other agent with probability $1/(1 + e^{-\beta \Delta f})$, where Δf is the other’s fitness minus its own and $\beta = 1$ per payoff unit. We assume rare mutation: each newcomer fixes or disappears before the next arrives. Pairwise fixation probabilities define a Markov chain over the five one-model populations. Its *stationary mass* gives each model’s long-run weight. With $N = 6$ the whole population is one table; with $N = 30$, fitness averages over random, potentially overlapping tables. Changing N changes both table compositions and fixation dynamics, not only whom an agent is compared with. We evaluate this specified finite-population process, not a scoring-only intervention.

Scoring against tablemates rewards the smallest payer. Within one game, Proposition 2 makes the ranking mechanical: the seat that pays less earns more. What the data add is who pays least in company, which is Qwen3-235B, and what this selection costs the group. Selection gives Qwen3-235B the largest weight at every risk level (Figure 6a). At $p = 0.5$, where reaching the target is one of two optima, and at $p = 0.9$, where it is the only one, the selected population reaches it far less often than a model picked at random (47% against 76% at $p = 0.9$), and stronger selection pushes group success close to zero. One game shows the mechanism. Beside five exact fair-share seats at $p = 0.9$, a seat that skips its last payment

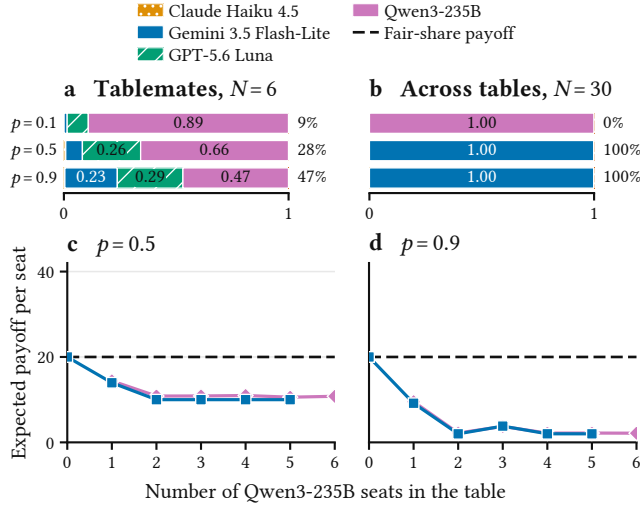


Figure 6: Rare-mutation Fermi selection ($\beta = 1$). (a, b) Stationary mass in a population of one table ($N = 6$) or a larger randomly mixed population ($N = 30$); right of each bar, the self-play target rate weighted by mass. (c, d) Expected payoff of a Qwen3-235B or Gemini 3.5 Flash-Lite seat by the number of Qwen3-235B seats in the same table, at $p = 0.5$ and $p = 0.9$; dashed: fair-share payoff. Invasion diagrams are in the supplement.

makes the table miss, so its expected payoff falls from 20 to 2.2, yet it still earns more than each tablemate, who expects 2. This is the known result that selection on payoff relative to a small group favours whoever does better than its neighbours, even at a cost to the group [23], here shown with the play of deployed language models.

A larger population favours the fair-share player. In the larger population, selection favours Gemini 3.5 Flash-Lite at moderate and high risk (Figure 6b). Its observed self-play attains the welfare optimum, but Nash equilibrium alone does not imply resistance to payoff-based imitation. The ranking comes from the estimated payoffs at every composition along the fixation path. In a mixed table, a lower payer can earn more than its tablemates even while lowering its own payoff (Figure 6c, d). At low risk both population sizes favour Qwen3-235B, still far below the optimum. The switch to Gemini 3.5 Flash-Lite occurs already at seven or eight agents. The winners persist at the stronger selection intensities and larger populations tested and with grid-only self-play payoffs; each stays on top in at least 93% of bootstrap samples. Under weak selection, within-table masses are near uniform. This population-size dependence is consistent with finite-population theory [23]. Which model wins remains conditional on the base-prompt payoffs, whose signatures change with wording. We do not establish the same ranking under other prompts or selection processes.

In post-hoc candidate-set checks, removing Qwen3-235B makes GPT-5.6 Luna the one-table winner at all tested risks. Removing Gemini 3.5 Flash-Lite makes GPT-5.6 Luna the larger-population winner at moderate and high risk. The supplement reports every

omission and the mass-weighted welfare, including at the $p = 0.5$ tie. These point estimates test candidate-set dependence, not robustness to a new prompt.

8 Discussion

Possible causes. Our data record choices, not reasoning, so we offer hypotheses. The equal split may be a focal point that models reconstruct without the hint [6]; a question may call up computation while a move request calls up a default [7, 24]; and post-training may favour doing one’s share when nothing says otherwise, consistent with models being more cooperative than people [16]. Stronger reasoning is not an obvious fix, since reasoning models can free-ride more in public goods games [10].

Consequences for design. State the agent’s goal, since naming it removed most zero-risk payment. Report $\text{opt}(p)$, a dominated-action condition and payment after settlement, since success can coexist with wasted money. Test mixed populations, where an exact fair share can fail beside one late dropout. Audit how population size and the comparison process affect selection; a tablemate payoff advantage need not improve group welfare.

Threats to validity. The goal sentence is itself an instruction, tested in one phrasing and in self-play only. A group-payoff goal is untested, so we cannot separate objective clarity from the direction of the instruction. The running pool is not printed; arithmetic errors may explain some payment after settlement. Value questions omit $p = 0$. Shared prompts can also affect models differently. These gaps limit causal explanations of the observed defaults.

Limitations. We test five low-cost models, one per provider, in English and in one game; claims concern this panel, not the whole tier. Ten games per cell limit power, and a zero-width bootstrap interval in a constant sample does not prove deterministic behaviour. Rewordings cover a small prompt space, welfare losses assume risk neutrality, and scripted, mixed and selection tests use the base prompt. Fixed seat blocks confound composition with position, and selection assumes rare entry. We will release prompts, configurations, seeds and code; they reproduce the design, not the model transcripts.

Ethics. The study involves no human subjects and no personal data. We report failure modes of deployed systems so that designers can test for them before trusting agents with shared resources.

References

- [1] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. 2025. Playing repeated games with large language models. *Nature Human Behaviour* 9, 7 (2025), 1380–1390. doi:10.1038/s41562-025-02172-y
- [2] James Andreoni. 1988. Why free ride? Strategies and learning in public goods experiments. *Journal of Public Economics* 37, 3 (1988), 291–304. doi:10.1016/0047-2727(88)90043-6
- [3] Luis Celiktemel, Edward Eichhorn, Levin Brinkmann, Robin Schimmelpfennig, Aron Vallinder, Yaomin Jiang, Edward Hughes, and Iyad Rahwan. 2026. Group Selection Promotes Prosocial Prompts in Populations of LLM Agents. arXiv:2606.23343 doi:10.48550/arXiv.2606.23343
- [4] Yiting Chen, Tracy Xiao Liu, You Shan, and Songfa Zhong. 2023. The emergence of economic rationality of GPT. *Proceedings of the National Academy of Sciences* 120, 51 (2023), e2316205120. doi:10.1073/pnas.2316205120
- [5] Allan Dafoe, Edward Hughes, Yoram Bachrach, Tantum Collins, Kevin R. McKee, Joel Z. Leibo, Kate Larson, and Thore Graepel. 2020. Open Problems in Cooperative AI. arXiv:2012.08630 doi:10.48550/arXiv.2012.08630
- [6] Éléonore Dodiviers and Ismaël Rafai. 2025. Uncovering the fairness of AI: exploring focal point, inequality aversion, and altruism in ChatGPT’s dictator game decisions. *Economics Bulletin* 45, 4 (2025), 1818–1825.
- [7] Caoyun Fan, Jindou Chen, Yaohui Jin, and Hao He. 2024. Can Large Language Models Serve as Rational Players in Game Theory? A Systematic Analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 17960–17967. doi:10.1609/aaai.v38i16.29751
- [8] Apostolos Filippas, John J. Horton, and Benjamin S. Manning. 2024. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?. In *Proceedings of the 25th ACM Conference on Economics and Computation (EC ’24)*. ACM, 614–615. doi:10.1145/3670865.3673513
- [9] Nicolò Fontana, Francesco Pierri, and Luca Maria Aiello. 2025. Nicer than Humans: How Do Large Language Models Behave in the Prisoner’s Dilemma? *Proceedings of the International AAAI Conference on Web and Social Media* 19 (2025), 522–535. doi:10.1609/icwsm.v19i1.35829
- [10] David Guzman Piedrahita, Yongjin Yang, Mrinmaya Sachan, Giorgia Ramponi, Bernhard Schölkopf, and Zhijiang Jin. 2025. Corrupted by Reasoning: Reasoning Language Models Become Free-Riders in Public Goods Games. In *Conference on Language Modeling (COLM)*. https://openreview.net/forum?id=kH6L0HGJEl
- [11] Jen-tse Huang, Eric John Li, Man Ho Lam, Tian Liang, Wenxuan Wang, Youliang Yuan, Wenxiang Jiao, Xing Wang, Zhaopeng Tu, and Michael R. Lyu. 2025. Competing Large Language Models in Multi-Agent Gaming Environments. In *International Conference on Learning Representations (ICLR)*. 65639–65681.
- [12] Edward Hughes, Joel Z. Leibo, Matthew Phillips, Karl Tuyls, Edgar Dueñez-Guzman, Antonio García Castañeda, Iain Dunning, Tina Zhu, Kevin McKee, Raphael Koster, Heather Roff, and Thore Graepel. 2018. Inequity aversion improves cooperation in intertemporal social dilemmas. In *Advances in Neural Information Processing Systems*, Vol. 31. Curran Associates, Inc.
- [13] Shivani Kumar, Adarsh Bharathwaj, and David Jurgens. 2026. Cooperative Profiles Predict Multi-Agent LLM Team Performance in AI for Science Workflows. arXiv:2604.20658 doi:10.48550/arXiv.2604.20658
- [14] Joel Z. Leibo, Edgar A. Dueñez-Guzman, Alexander Vezhnevets, John P. Agapiou, Peter Sunehag, Raphael Koster, Jayd Matyas, Charlie Beattie, Igor Mordatch, and Thore Graepel. 2021. Scalable Evaluation of Multi-Agent Reinforcement Learning with Melting Pot. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)*. PMLR, 6187–6199. https://proceedings.mlr.press/v139/leibo21a.html
- [15] Joel Z. Leibo, Vinicius Zambaldi, Marc Lanctot, Janusz Marecki, and Thore Graepel. 2017. Multi-agent Reinforcement Learning in Sequential Social Dilemmas. In *Proceedings of the 16th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2017)*. IFAAMAS, 464–473. https://www.ifaamas.org/Proceedings/aamas2017/pdfs/p464.pdf
- [16] Qiaozhu Mei, Yutong Xie, Walter Yuan, and Matthew O. Jackson. 2024. A Turing test of whether AI chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences* 121, 9 (2024), e2313925121. doi:10.1073/pnas.2313925121
- [17] Manfred Milinski, Ralf D. Sommerfeld, Hans-Jürgen Krambeck, Floyd A. Reed, and Jochem Marotzke. 2008. The collective-risk social dilemma and the prevention of simulated dangerous climate change. *Proceedings of the National Academy of Sciences* 105, 7 (2008), 2291–2294. doi:10.1073/pnas.0709546105
- [18] Shayegan Omidshafiei, Christos Papadimitriou, Georgios Piliouras, Karl Tuyls, Mark Rowland, Jean-Baptiste Lespiau, Wojciech M. Czarnecki, Marc Lanctot, Julien Perolat, and Remi Munos. 2019. α -Rank: Multi-Agent Evaluation by Evolution. *Scientific Reports* 9, 1 (2019), 9937. doi:10.1038/s41598-019-45619-9
- [19] Thomas R. Palfrey and Howard Rosenthal. 1984. Participation and the provision of discrete public goods: a strategic analysis. *Journal of Public Economics* 24, 2 (1984), 171–193. doi:10.1016/0047-2727(84)90023-9
- [20] Kenneth Payne and Baptiste Alloui-Cros. 2025. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory. arXiv:2507.02618 doi:10.48550/arXiv.2507.02618
- [21] Giorgio Piatti, Zhijiang Jin, Max Kleiman-Weiner, Bernhard Schölkopf, Mrinmaya Sachan, and Rada Mihalcea. 2024. Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents. In *Advances in Neural Information Processing Systems*, Vol. 37. Curran Associates, Inc., 111715–111759. doi:10.52202/079017-3548
- [22] Francisco C. Santos and Jorge M. Pacheco. 2011. Risk of collective failure provides an escape from the tragedy of the commons. *Proceedings of the National Academy of Sciences* 108, 26 (2011), 10421–10425. doi:10.1073/pnas.1015648108
- [23] Mark E. Schaffer. 1988. Evolutionarily stable strategies for a finite population and a variable contest size. *Journal of Theoretical Biology* 132, 4 (1988), 469–478. doi:10.1016/S0022-5193(88)80085-7
- [24] Thomas Schmied, Jörg Bornschein, Jordi Grau-Moya, Markus Wulfmeier, and Razvan Pascanu. 2026. LLMs are Greedy Agents: Effects of RL Fine-tuning on Decision-Making Abilities. In *International Conference on Learning Representations (ICLR)*. 127538–127571. https://openreview.net/forum?id=weUP6H5Ko9
- [25] Oliver Slumbers, Joel Z. Leibo, and Marco A. Janssen. 2025. Using Large Language Models to Simulate Human Behavioural Experiments: Port of Mars. arXiv:2506.05555 doi:10.48550/arXiv.2506.05555
- [26] Peter Stone, Gal A. Kaminka, Sarit Kraus, and Jeffrey S. Rosenschein. 2010. Ad Hoc Autonomous Agent Teams: Collaboration without Pre-Coordination. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 24. 1504–1509. doi:10.1609/aaai.v24i1.7529
- [27] Alessandro Tavoni, Astrid Dannenberg, Giorgos Kallis, and Andreas Löschel. 2011. Inequality, communication, and the avoidance of disastrous climate change in a public goods game. *Proceedings of the National Academy of Sciences* 108, 29 (2011), 11825–11829. doi:10.1073/pnas.1102493108
- [28] Inês Terrucha, Elias Fernández Domingos, Francisco C. Santos, Pieter Simoons, and Tom Lenaerts. 2024. The art of compensation: How hybrid teams solve collective-risk dilemmas. *PLOS ONE* 19, 2 (2024), e0297213. doi:10.1371/journal.pone.0297213
- [29] Emanuel Tewolde, Xiao Zhang, David Guzman Piedrahita, Vincent Conitzer, and Zhijiang Jin. 2026. CoopEval: Benchmarking Cooperation-Sustaining Mechanisms and LLM Agents in Social Dilemmas. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/2604.15267
- [30] Arne Traulsen and Martin A. Nowak. 2006. Evolution of cooperation by multilevel selection. *Proceedings of the National Academy of Sciences* 103, 29 (2006), 10952–10955. doi:10.1073/pnas.0602530103
- [31] Karl Tuyls, Julien Perolat, Marc Lanctot, Joel Z. Leibo, and Thore Graepel. 2018. A Generalised Method for Empirical Game-Theoretic Analysis. In *Proceedings of the 17th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2018)*. IFAAMAS, 77–85. https://ifaamas.org/Proceedings/aamas2018/pdfs/p77.pdf
- [32] Aron Vallinder and Edward Hughes. 2025. Cultural Evolution of Cooperation among LLM Agents: Extended Abstract. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)*. IFAAMAS, 2771–2773. https://www.ifaamas.org/Proceedings/aamas2025/pdfs/p2771.pdf
- [33] Michael P. Wellman. 2006. Methods for Empirical Game-Theoretic Analysis (Extended Abstract). In *Proceedings of the Twenty-First National Conference on Artificial Intelligence (AAAI-06)*. AAAI Press, 1552–1555.
- [34] Richard Willis, Jianing Zhao, Yali Du, and Joel Z. Leibo. 2026. Evaluating Collective Behaviour of Hundreds of LLM Agents. Strategic Engineering Workshop on LLMs and Game Theory (SE@AAMAS 2026). arXiv:2602.16662 doi:10.48550/arXiv.2602.16662